BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/BEE 4750/hw2-regina`
```

```julia
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```julia
"""
                    (+)
Atmospheric CO         Global Mean Temperature

                                 (-)
                                |
                           Ocean CO

        (-)        -
"""
```

```
"                    (+)\nAtmospheric CO         Global Mean Temperature\n
```

Problem 1 SOLUTION:

Henry's Law says that solubility of CO in seawater decreases as temperature increases. In other words, warm water holds less CO. Because of this, the arrow pointing from global mean temperature to the concentration of CO in the ocean is a negative (-) sign and this is a dampening relationship. As temperature increases, the ocean's water warms and CO concentrations decrease.

When concentrations of CO in the ocean increase, the concentration of CO in the atmosphere decrease since the ocean is acting as a carbon sink and taking out CO from the atmosphere. Because of this, the sign is negative (-) and this is a dampening relationship.

Finally, when CO concentrations in the atmosphere increases, the greenhouse effect is magnified since the CO is now absorbing heat and re-emitting it. As a result, the global mean temperature to increase. This is an amplifying relationship and the arrow's sign is positive (+).

This suggests that the overall feedback between temperature and the ocean carbon cycle is an amplyfing (or positive) feedback because an increase in atmospheric CO , increases temperature, which creates less CO2 in the ocean, which means less CO is being absorbed by the ocean and there is more of it in the atmosphere, which finally leads to even more warming.

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

2.1 Solution:

```
          |
          |
   -----|-----
  |           |
--->        ---->
  |            |
```

Two boxes or control volumes are needed for this because there are two input streams, one before X = 15 km and one after. These streams are causing a change in flow rate of the river so the control volumes will help model and understand those changes. Two equations are needed; one to model the first discharge and another to model the second discharge. This is because the initial conditions are different - the river inflow before the first waste stream will have changed due to a few factors such as decay. This will have created a new initial condition of the water mixing with the second waste stream.

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

```
Q = 250000 # river flow rate (m^3/d)
Ci = 0.5/1000 # initial CRUD concentration in river (kg/m^3)
Q1 = 40000 # first wastewater input (kg/m^3)
C1 = 9/1000 # concentration in first wastewater input (kg/m^3)
Q2 = 60000 # second WW input (kg/m^3)
C2 = 7/1000 # concentration in second WW input (kg/m^3)

k = 0.36 # first order decay rate (1/day)
d = 54 # rate CRUD is deposited by the atmosphere along the river (kg/km/d)
v = 10 # constant average velocity of the river (km/day)
L = 15 # distance between WW input 1 and 2 (km)

# fcn to find concentration at steady state
function steady_state_C(Q)
    # dC/dx = -decay rate + deposit rate = (-kC/v) + d/Q when dC/dx = 0 because steady state
    k = 0.36 # first order decay rate (1/day)
    d = 54 # rate CRUD is deposited by the atmosphere along the river (kg/km/d)
    v = 10 # constant average velocity of the river (km/day)
    return((v*d)/(k*Q))
end

# fcn to find the flow and concentration output of two streams mixing
function two_streams_mix(Q1, C1, Q2, C2)
    Q_out = Q1 + Q2
    C_out = ((Q1*C1) + (Q2*C2)) / Q_out
    return (Q_out, C_out)
end

function conc_model(x, C0, Q)
    k = 0.36 # first order decay rate (1/day)
    d = 54 # rate CRUD is deposited by the atmosphere along the river (kg/km/d)
    v = 10 # constant average velocity of the river (km/day)
    C_out = steady_state_C(Q) + (C0 - steady_state_C(Q)) * exp.((-k/v) * (x))
```

```
    return(C_out)
end

# FIRST CONTROL VOLUME: 0 <= x < L = 15 km
(Q3, C3) = two_streams_mix(Q, Ci, Q1, C1)
C_L = conc_model(L, C3, Q3)
"""
x1 = range(0.0, L; length=200)
C1 = conc_model(x1, 0.0, C3, Q3)
C_L = last(C1) # concentration at L = 15 km
"""

# SECOND CONTROL VOLUME: L = 15 km >= x
(Q4, C4) = two_streams_mix(Q3, C_L, Q2, C2)
C_down_stream = conc_model(20, C4, Q4)

"""
x2 = range(L, 60; length=400)
C1 = conc_model(x2, L, C_L, Q4)
"""

println("C(x = 0 km) after wastewater input #1  = ", round(C3*1000, digits=3), " kg/(1000m^3
println("C(x = 15 km) right before wastewater input #2 = ", round(C_L*1000, digits=3), " kg/
println("C(x = 20 km ) after wastewater input #2  = ", round(C4*1000, digits=3), " kg/(1000m
```

```
C(x = 0 km) after wastewater input #1  = 1.672 kg/(1000m^3)
C(x = 15 km) right before wastewater input #2 = 3.133 kg/(1000m^3)
C(x = 20 km ) after wastewater input #2  = 3.796 kg/(1000m^3)
```

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

By observing the values of $C(0+)$, $C(L-)$, and $C(L+)$, we can tell the system is not in compliance with the regulatory standard of 2.3 kg/(1000m^3). The stream after waste input 1 has a concentration of 1.7 kg/(1000m^3), which is within the regulatory standards. However, after traveling downstream, the waste concentration in the stream continues to increase, causing the concentration to rise to 3.1 kg/(1000m^3) and 3.9 kg/(1000m^3) after the second source of contamination - both above the regulatory standard of 2.3 kg/(1000m^3).

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} \underbrace{+ a \ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Remove the greenhouse term, since it is not relevant in pre-industrial applications:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}}$$

Divide by C to isolate $\frac{dT}{dt}$ and distribute the negative:

$$\frac{dT}{dt} \;=\; \frac{1}{C}\left(\frac{(1-\alpha)S}{4} - A + B\,T\right) \tag{1}$$

Discretize the formula using the forward Euler method of the time derivative at t:

$$\frac{T(t+\Delta t) - T(t)}{\Delta t} \;=\; \frac{1}{C}\left(\frac{(1-\alpha)S}{4} - A + B\,T(t)\right) \tag{2}$$

Multiply by Δt and add $T(t)$ to the RHS to simplify the discretized differential equation:

$$T(t+\Delta t) \;=\; T(t) + \frac{\Delta t}{C}\left(\frac{(1-\alpha)S}{4} - A + B\,T(t)\right) \tag{3}$$

Problem 3.2

Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^{\circ}\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^{\circ}\text{C} \leq T \leq 10^{\circ}\text{C} \\ \alpha_0 & \text{if } T \geq 10^{\circ}\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^{\circ}\text{C} \leq T_0 \leq 30^{\circ}\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

References

List any external references consulted, including classmates.